Antiderivatives & Differential Equations

MATH 145 – Applied Calculus

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November 1, 2024

Inverse Operations

In a previous lecture, we talked about *inverse functions* and *inverse operations*.

For example, if I add 4 to a number, I can subtract 4 from the answer to reverse or "undo" the addition, as subtraction is the inverse operation to addition.

Or if I take e to the fourth power, e^4, if I want to reverse that or "undo" it, I can take the natural logarithm, which is the inverse operation of exponentiation.

If we think of differentiation as an "operation" mapping one function f to its derivative f', then what would be the equivalent "inverse operation"?

Antiderivatives

If $f(x)$ is a function, then an **antiderivative** of $f(x)$ is a function $F(x)$ such that $F'(x) = f(x)$.

Antiderivatives

Example: Find an antiderivative for $f(x) = 2x$.

Integral Notation

Antiderivatives

If $f(x)$ is a function, then the most general antiderivative of $f(x)$ is denoted by:

$$\int f(x)\,dx$$

Example:

$$\int 2x\,dx = x^2 + C$$

Rules for Antiderivatives

Rules for Finding Antiderivatives:

- (Inverse) Power Rule:

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \text{ for all } n \neq 1.$$

- Sum Rule:

$$\int \left(f(x) + g(x)\right) \, dx = \int f(x) \, dx + \int g(x) \, dx.$$

- Constant Multiple Rule:

$$\int c f(x) \, dx = c \int f(x) \, dx.$$

Example: Antiderivatives

Example: Find $\displaystyle\int 5x + 9\,dx$.

Example: Find $\displaystyle\int 2x^2 - 3x + 7\,dx$.

Example: Find $\displaystyle\int \frac{(x+1)(x-2)}{x^4}\,dx$.

Example: If $F'(x) = 6x^3$, find $F(x)$.

Example: If $F'(x) = 6x^3$, find $F(x)$ such that $F(1) = 3$.

Example: A car is traveling at the rate of $88\,{}^{ft}/_{sec}$ when the brakes are applied. The car begins decelerating at a constant rate of $15\,{}^{ft}/_{sec^2}$.

1. How many seconds before the car stops?
2. How far does the car travel during that time?

Example: The velocity function is $v(t) = t^2 - 3t + 2$ for a particle moving along a line. Find the displacement (net distance covered) of the particle during the time interval $[-2, 5]$.